

3.3. LONG-RUN RISK

Real consumption growth, which we observe in data, has an autocorrelation close to zero. The data in Table 8.3, for example, exhibit a quarterly autocorrelation of 0.08. The basic Mehra–Prescott model assumes that the consumption shocks are independent and identically distributed (i.i.d.), or that they cannot be forecasted. In an influential article, Bansal and Yaron (2004) change this assumption to specifying that the mean of consumption growth is not constant but wanders around so slowly that it is extremely hard to distinguish between an i.i.d. process and this slow, persistent process where there is a small amount of forecastability.⁷ This changing consumption process is what Bansal and Yaron call long-run risk.⁸ Bansal and Yaron also alter the basic CRRA preference in the Mehra–Prescott framework so that the representative agent cares much more about long-run risks.⁹ The Bansal–Yaron long-run risk model has been adapted to many different contexts and explains a wide variety of stylized facts including the equity premium puzzle and the risk-free rate puzzle of this section.¹⁰

Europe at the time of writing is a wonderful example of the long-run risk story at play. Current shocks, say, from a particular raucous parliamentary vote in some euro zone country, have an effect on today's expectation about future economic growth. But they also have an effect on economic growth ten to twenty years in the future. The vote could diminish current productivity, but lowering current productivity might keep Europe on a low-growth path for a long time. Long-run risks exist. The special preferences Bansal and Yaron employ differentiate between the short-run and the long-run effect of these shocks. Equities are particularly sensitive to long-run risk, not surprisingly, because equity is a long-lived (in fact, perpetual) security.

The second channel that Bansal and Yaron employ is that the fundamental consumption factor exhibits time-varying volatility. Agents dislike volatility components and asset prices reflect the volatility risk. As volatility increases, asset valuation declines. The Bansal–Yaron model has separate risk compensations for consumption growth and for consumption volatility.

⁷ Bansal and Yaron (2004) also specify that the volatility of the conditional mean of consumption varies over time.

⁸ Predictability in consumption growth had been considered before, but allowing for simple autocorrelated consumption growth actually makes the equity premium puzzle worse with CRRA utility because it smooths out consumption, requiring even higher degrees of risk aversion to match the data. See Dunn and Singleton (1986) for an early attempt.

⁹ Specifically, the representative agent wishes to have smooth consumption paths, but the way that these consumption paths are smoothed is different from risk aversion, which measures how an agent treats different risky betas at a given moment in time. These are called Epstein and Zin (1989) preferences, and they are also used in the disaster framework.

¹⁰ See Bansal (2007) for a summary of this literature that was already large by then.